

Geometry: Proportion Properties (Ratio & Similarity)



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DIRECTIONS

Apply the named proportion property to complete or verify each statement. Show all work.

- 1 Means-Extremes Property. Find x:

$$\frac{6}{4} = \frac{x}{5}$$

Answer: _____

- 2 Alternation Property. If the proportion holds, rewrite using alternation:

$$\frac{4}{4} = \frac{8}{8} \Rightarrow \frac{4}{8} = \dots$$

Answer: _____

- 3 Inverse Property. Write the inverse proportion:

$$\frac{7}{7} = \frac{28}{28} \Rightarrow \frac{7}{28} = \dots$$

Answer: _____

- 4 Addition Property (Componendo). Complete:

$$\frac{8}{4} = \frac{16}{8} \Rightarrow \frac{x}{4} = \frac{24}{8}$$

Answer: _____

- 5 Subtraction Property (Dividendo). Complete:

$$\frac{9}{2} = \frac{18}{4} \Rightarrow \frac{x}{2} = \frac{14}{4}$$

Answer: _____

- 6 Componendo-Dividendo. Verify the property:

$$\frac{7}{3} = \frac{21}{9} \Rightarrow \frac{10}{4} = \frac{30}{12}$$

Answer: _____

- 7 Extended ratio. Find all three parts:

Parts in ratio 4 : 2 : 1, Total = 28

Answer: _____

- 8 Cross-Products. Solve for x:

$$\frac{3}{4} = \frac{x+6}{16}$$

Answer: _____

- 9 Proportion Property for similar figures. Find x:

$$\frac{4}{16} = \frac{5}{x}$$

Answer: _____

- 10 Name the proportion property shown:

$$\frac{2}{3} = \frac{6}{9} \Rightarrow \frac{5}{3} = \frac{15}{9}$$

Answer: _____



Answer Key & Solutions

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TEACHER NOTES

Cover: Means-Extremes (cross products), Alternation, Inverse, Componendo (Addition), Dividendo (Subtraction), Componendo-Dividendo, and extended ratios.

1 Means-Extremes Property. Find x:

$$\frac{6}{4} = \frac{x}{5}$$

$$= x = \frac{15}{2}$$

Cross multiply: $6x = 4 \times 5 = 20$, $x = 20/6$.

2 Alternation Property. If the proportion holds, rewrite using alternation:

$$\frac{4}{4} = \frac{8}{8} \Rightarrow \frac{4}{8} = \dots$$

$$= \frac{4}{8}$$

Alternation: swap means. $4/4 = 8/8 \rightarrow 4/8 = 4/8$.

3 Inverse Property. Write the inverse proportion:

$$\frac{7}{7} = \frac{28}{28} \Rightarrow \frac{7}{7} = \dots$$

$$= \frac{28}{28}$$

Inverse: flip both fractions. $b/a = d/c \rightarrow 7/7 = 28/28$.

4 Addition Property (Componendo). Complete:

$$\frac{8}{4} = \frac{16}{8} \Rightarrow \frac{x}{4} = \frac{24}{8}$$

$$= x = 12$$

Componendo: add denominator to numerator. $(a+b)/b = (c+d)/d \rightarrow (8+4)/4 = (16+8)/8$.

5 Subtraction Property (Dividendo). Complete:

$$\frac{9}{2} = \frac{18}{4} \Rightarrow \frac{x}{2} = \frac{14}{4}$$

$$= x = 7$$

Dividendo: subtract denominator from numerator. $(a-b)/b = (c-d)/d \rightarrow (9-2)/2 = (18-4)/4$.

6 Componendo-Dividendo. Verify the property:

$$\frac{7}{3} = \frac{21}{9} \Rightarrow \frac{10}{4} = \frac{30}{12}$$

$$= \text{True: } 10 \cdot 12 = 30 \cdot 4 = 120$$

Check: $10 \times 12 = 120$ and $30 \times 4 = 120$. Equal ✓

7 Extended ratio. Find all three parts:

Parts in ratio 4 : 2 : 1, Total = 28

$$= 16, 8, 4$$

Unit = $28 \div (4+2+1) = 4$. Parts: $4 \times 4 = 16$, $2 \times 4 = 8$, $1 \times 4 = 4$.

8 Cross-Products. Solve for x:

$$\frac{3}{4} = \frac{x+6}{16}$$

$$= x = 6$$

Cross multiply: $4(x+6) = 3 \times 16 = 48$. So $4x = 48 - 24 = 24$, $x = 6$.

9 Proportion Property for similar figures. Find x:

$$\frac{4}{16} = \frac{5}{x}$$

$$= x = 20$$

Scale factor = 4. Cross multiply: $4x = 5 \times 16 = 80$. $x = 80 \div 4 = 20$.

10 Name the proportion property shown:

$$\frac{2}{3} = \frac{6}{9} \Rightarrow \frac{5}{3} = \frac{15}{9}$$

$$= \text{Componendo (Addition)}$$

Add denominator to numerator on both sides.

